

Aurorae

Spectacular displays of colorful, shimmering lights in the night sky

The aurora borealis, or northern lights (in the Northern Hemisphere), and the aurora australis, or southern lights (in the Southern Hemisphere), appear at latitudes greater than 60° North and South, respectively. They are centered on Earth's magnetic poles and occur when electrically charged particles from the Sun are funneled into Earth's magnetic field above the poles and collide with atoms in the upper atmosphere to trigger reactions that produce streams of light (see panel, right). Some displays are almost static patches of diffuse brightness, but others feature constantly moving lights suggestive of curtains waving in a breeze. Every fold of the curtain consists of many parallel rays, each lined up with the local direction of Earth's magnetic field. The color of an aurora depends on its height and which gas in the atmosphere is being excited. Green is the most common color, but displays of red, blue, violet, or even pink occur occasionally.

Cloudless conditions away from light pollution during the long nights of winter are ideal for watching aurorae. Alaska, Canada, northern Scandinavia, and northern Russia are the best places to see the northern lights. The southern lights can be seen from southern South America, Tasmania, and New Zealand's South Island, although near-inaccessible Antarctica has the best shows.

The **lights** of an aurora are **produced** mostly **56–90 miles (90–150 km)** above **Earth**, although very high (red) displays may be **up to 620 miles (1,000 km)** away



△ CURTAIN OF COLOR

The dazzling display of an aurora can be seen reflected in the waters of the Jökulsárlón glacial lagoon in Iceland.

▷ A VIEW FROM SPACE

Astronauts orbiting in the International Space Station have a unique view of the aurora borealis.

